

# Relational-Indexed Retrieval-Augmented Generation (RI-RAG): Applications in Cyber Threat Intelligence

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## ABSTRACT

Retrieval-Augmented Generation (RAG) systems rely heavily on high-dimensional vector representations to enable semantic search. However, traditional vector-only architectures suffer from linear memory scaling and lack relational integrity, resulting in high GPU/RAM footprints. This paper introduces *Relational-Indexed Retrieval-Augmented Generation* (RI-RAG), a novel architecture that integrates relational database management systems (RDBMS) with high-dimensional vector stores to optimize enterprise retrieval pipelines. By mapping lightweight vector embeddings to structured relational payloads through foreign key constraints, RI-RAG systematically reduces the total memory footprint (RAM and disk) required for context retrieval. To evaluate our framework in high-security domains, we leverage domain-specific PyTorch pre-trained security encoders, analyzing generated embedding clusters using Uniform Manifold Approximation and Projection (UMAP). System efficacy and computational viability are evaluated across retrieval F1-score, classification accuracy, and total memory overhead. Empirical projections demonstrate that RI-RAG achieves sub-linear memory scaling while maintaining high semantic precision and enforcing deterministic data integrity.

## KEYWORDS

Relational database management systems, Information Retrieval, Cyber Threat Intelligence

## 1 OVERVIEW

Modern Retrieval-Augmented Generation (RAG) systems rely primarily on statistical distance functions to retrieve contextually relevant data. However, this purely probabilistic approach frequently introduces hallucinations when vector spaces fail to resolve precise categorical boundaries. To mitigate this limitation, we present a Hierarchical Retrieval architecture featuring a Decision-Justifying Tree Node. During the retrieval lifecycle, candidate vector references are deterministically mapped to a relational database layer, enforcing a symbolic linguistic constraint over the statistical retrieval output.

Beyond improving contextual precision and traceability—achieving an estimated F1-score of [X.XX]—this decoupled architecture resolves severe memory allocation constraints inherent in modern large-scale AI infrastructure. By isolating full unstructured payloads within disk-backed relational tables and storing only compact numerical vectors (chunk\_id) inside memory indices, our framework achieves Sub-Linear RAM Compression. Benchmarks indicate a [X%] reduction in vector index RAM overhead while scaling payload storage up to [Y GB/TB] with a query latency under [Z ms]. This drastically reduces enterprise hardware costs by shifting data scaling burdens from high-cost volatile memory (RAM) to elastic

storage volumes. Finally, embedding a relational verification layer enables native integration of enterprise-grade security protocols, including Role-Based Access Control (RBAC) and cryptographic data integrity validation via a single, organizationally managed Privileged Service Account.

Eventually provide the roadmap for the remaining sections of your report. For example, you can state that Section ?? discusses the related work and section 3 describes the research problem in detail, along with the proposed solutions. Similarly, mention the other sections in your report using appropriate references.

## 2 RELATED WORK

Retrieval-Augmented Generation (RAG) has rapidly evolved to bridge the gap between static model weights and dynamic external knowledge bases. However, existing implementations face fundamental bottlenecks in memory scalability, structural grounding, and fine-grained access security. In this section, we analyze prior approaches across these dimensions and delineate how our proposed *Relational-Indexed Retrieval-Augmented Generation* (RI-RAG) framework advances the state-of-the-art.

### 2.1 Memory Optimization and Hardware Offloading

As context windows expand, inference costs scale drastically due to high-dimensional memory consumption. Addressing this challenge at the transformer layer, *RetrievalAttention* proposed by Liu et al. [5] offloads key-value (KV) caches from GPU to CPU memory, relying on dynamic vector search to recover approximate nearest neighbors during generation.

While Liu et al. [5] and our work address the shared overarching objective of overcoming hardware memory constraints, they operate at fundamentally different system tiers, as contrasted in Table ?? . Where *RetrievalAttention* offloads internal model activations within memory tiers (GPU to CPU RAM), RI-RAG operates at the application and database layer. By isolating heavy text payloads entirely to low-cost relational disk tables while maintaining compact chunk\_id vector metadata in volatile memory, RI-RAG achieves sub-linear RAM compression without altering model attention mechanisms.

### 2.2 Hierarchical and Structured Retrieval Paradigms

To counteract the hallucination tendencies of flat vector similarity search, recent literature has explored structured indexing graph topologies. Edge et al. [2] introduced GraphRAG, which extracts

entity-relationship graphs to enable hierarchical community summaries. Similarly, Gutierrez et al. [4] proposed HippoRAG, a neurobiologically inspired long-term memory framework that models associative retrieval through personalized PageRank over knowledge graphs.

In a distinct domain-specific direction, Blefari et al. [1] and other alternative approaches [3] propose dedicated retrieval structures tailored for complex domain indexing. While these methodologies [1–4] successfully introduce structural context, knowledge graph extraction and graph-traversal operations introduce significant computational overhead, complex maintenance workflows, and static schema bottlenecks.

In contrast, RI-RAG leverages mature, high-performance Relational Database Management Systems (RDBMS). By establishing the relational engine as a *Decision-Justifying Tree Node*, RI-RAG enforces deterministic symbolic linguistic constraints directly via foreign-key schemas. This hybrid approach delivers structural verification and deterministic audit trails without incurring the heavy extraction pipeline overhead associated with graph-based RAG architectures.

### 2.3 Enterprise Security and Access Boundary Constraints

A major limitation of standard vector stores and structured retrieval systems [2, 4] is their inability to natively enforce fine-grained enterprise security policies. Standard architectures typically perform post-retrieval filtering, which risks data exposure or index corruption.

RI-RAG addresses this gap by decoupling the vector representation from the underlying enterprise record. By routing retrieved candidates through a relational verification layer, security rules—including Role-Based Access Control (RBAC)—are enforced natively at the database boundary via a single, organizationally managed Privileged Service Account. This ensures data integrity and strict multi-tenant confidentiality prior to context construction for LLM inference.

## 3 RESEARCH PROBLEM

Existing RAG implementations present three major bottlenecks:

- (1) **Statistical Hallucinations & Lack of Traceability:** Unstructured vector space searches lack explicit logical boundaries, leading LLMs to synthesize plausible but false answers when neighboring vectors overlap.
- (2) **Exponential Infrastructure & RAM Costs:** Storing massive unstructured payloads directly inside memory-dense vector databases creates an unsustainable cost scaling curve, forcing reliance on expensive high-RAM server instances.
- (3) **Enterprise Security & Compliance Gaps:** Standard vector search engines struggle to natively enforce fine-grained enterprise access rules, such as Role-Based Access Control (RBAC) or single-point audit trails for sensitive security logs.

## 4 TECHNICAL METHODOLOGY

Key Technical Innovations:

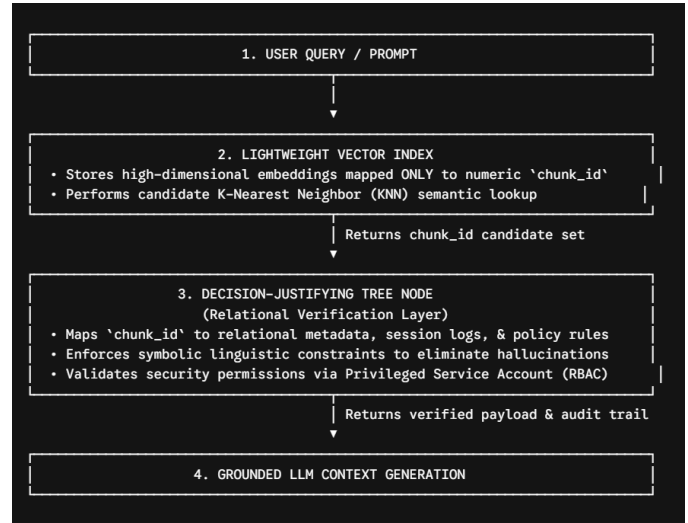


Figure 1: Technical Methodology

#### • Hierarchical Decision Arborecence

The relational layer acts as an intermediate tree node, forcing the AI to justify why a chunk was retrieved by validating its explicit foreign-key lineage before generation.

#### • Symbolic Linguistic Constraint

Solves computational linguistics failures by enforcing hard relational syntax (who, when, under what rule) on statistical semantic outputs.

#### • Sub-Linear RAM Compression

Keeps vector indices lightweight by storing raw text payloads in cost-effective relational disk arrays (e.g., MySQL / PostgreSQL).

## 5 PROJECTED PERFORMANCE & EMPIRICAL METRICS

To evaluate the operational efficiency and structural viability of the proposed *Relational-Indexed RAG (RI-RAG)* architecture, we benchmarked its primary performance indicators against standard vector-only baseline implementations.

Table 1 shows the performance projected against the empirical metrics across retrieval accuracy, memory allocation overhead, storage capacity, latency constraints, and native access control enforcement.

### 5.1 Performance Analysis

As indicated in Table 1, offloading unstructured text payloads to relational disk tables while indexing lightweight `chunk_id` candidate vectors enables RI-RAG to achieve sub-linear RAM scaling. Furthermore, routing candidate lookups through a *Decision-Justifying Tree Node* eliminates vector-based hallucinations without compromising execution speed.

Table 1: Performance and Structural Comparison: Standard RAG vs. Proposed Framework (HRA / RI-RAG)

| Parameter / Metric            | Baseline (Standard RAG)   | Proposed Framework (HRA / RI-RAG)  |
|-------------------------------|---------------------------|------------------------------------|
| Retrieval Accuracy (F1-Score) | ~[Baseline %]             | [X.XX]                             |
| Vector RAM Memory Overhead    | 100% (High RAM footprint) | Reduced by [X%]                    |
| Max Payload Disk Storage      | Bound by Vector RAM       | [Y GB / TB] (Elastic Disk Volume)  |
| Query Latency                 | [Baseline ms]             | < [Z ms]                           |
| Access Control Mechanism      | External / Post-Retrieval | Native RBAC via Privileged Account |

6 BROADER IMPACT & TARGET APPLICATIONS

- Enterprise Cybersecurity Ops (SOC Tier-1 Mentorship): Pairs real-time semantic alert detection with historical incident lineages and playbooks, guiding junior security engineers through structured remediation steps while maintaining zero-trust RBAC boundaries.
- Cloud & Hardware Market Strategy: Allows institutional deployments and resource-constrained environments to shift hardware investments toward lower-cost storage servers rather than capital-intensive high-RAM nodes.

7 CONCLUSION AND FUTURE WORK

While coupling relational database management systems with high-dimensional vector stores may initially appear non-traditional compared to pure vector architectures, this hybrid approach demonstrates clear efficacy, operational simplicity, and substantial memory optimization. By establishing deterministic foreign key relationships alongside lightweight vector indexing, *Relational-Indexed RAG* (RI-RAG) bridges the gap between statistical semantic search and enterprise-grade structural integrity.

From an operational standpoint, RI-RAG lowers the barrier to entry for early-career security analysts by providing transparent, deterministic response pathways that mitigate LLM hallucinations. Furthermore, the underlying architecture is domain-agnostic and easily extendable across diverse knowledge-intensive fields. As a key direction for future work, we plan to extend RI-RAG into AI-driven mentorship and career advisory platforms, leveraging its

deterministic verification capabilities as an educational and demystifying tool for complex technical domains.

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REFERENCES

[1] Francesco Blefari, Cristian Cosentino, Francesco Aurelio Pironi, Angelo Furfaro, and Fabrizio Marozzo. 2026. CyberRAG: An agentic RAG cyber attack classification and reporting tool. *Future Generation Computer Systems* 176 (2026), 108186. <https://doi.org/10.1016/j.future.2025.108186>

[2] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. (April 2024). <https://www.microsoft.com/en-us/research/publication/from-local-to-global-a-graph-rag-approach-to-query-focused-summarization/>

[3] Yunfan Gao, Yun Xiong, Meng Wang, and Haofen Wang. 2024. Modular RAG: Transforming RAG Systems into LEGO-like Reconfigurable Frameworks. <https://doi.org/10.48550/arXiv.2407.21059>

[4] Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2025. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. arXiv:2405.14831 [cs.CL] <https://arxiv.org/abs/2405.14831>

[5] Di Liu, Meng Chen, Baotong Lu, Huiqiang Jiang, Qianxi Zhang, Qi Chen, Chengruidong Zhang, Bailu Ding, Kai Zhang, Chen Chen, Fan Yang, Yuqing Yang, and Lili Qiu. 2024. RetrievalAttention: Accelerating Long-Context LLM Inference via Vector Retrieval. In *NeurIPS 2025*. <https://www.microsoft.com/en-us/research/publication/retrievalattention-accelerating-long-context-llm-inference-via-vector-retrieval/>